IZTECH – Department of Electrical and Electronics Engineering

EE331 – SIGNALS AND SYSTEMS

FINAL ASSIGNMENT REPORT

BANDPASS FILTER DESIGN

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Introduction

The aim of the homework was to design a digital bandpass filter that isolates a selected note of a recording containing eight distinct notes while attenuating the others and then analyze the filter's effect in both time and frequency domains.

Data Collection

The chosen audio "C_Major_Scale_2_unfiltered.wav" contains the C Major Scale played by a piano. The audio originates from a YouTube video, which was played on an iPad (9th Generation) tablet, and then recorded by an iPhone 13 cell phone in a quiet room. During recording, the phone was approximately 2 cm away from the tablet's speaker.

Method

1) Time – Domain Inspection and Audio Segmentation

The recording was read by audioread() function, which returned the signal samples x(t) and sampling frequency Fs. Then, the full waveform was plotted, where it was observed that notes appeared with high amplitudes that began to diminish as the note kept going on – until the next note, where another jump in amplitude was observed. Based on these amplitude transitions, the audio was manually segmented into eight time intervals, where each segment corresponded to one note of the C major scale.

2) Frequency – Domain Analysis and Fundamental Frequency Estimation

A frequency search range of 50–2000 Hz was selected to estimate the fundamental frequency of each note while avoiding DC and irrelevant high-frequency content. For each segmented note, the corresponding samples were extracted, the mean was removed, and a Hann window was applied to reduce spectral leakage. The magnitude spectrum was then obtained using the FFT, with fftshift() for a centered frequency axis. Finally, the fundamental frequency f0 was estimated as the dominant peak within the 50–2000 Hz band and stored for each note.

3) Filter Design & Application

A Butterworth filter was chosen as the filter type, due to the Butterworth filter providing smooth passband while having an average/moderate order. First, the target note was selected, and the target fundamental frequency was determined from target note. The passband was chosen as:

$$[f0 - \Delta f\,, f0 + \Delta f]$$

The value 15 Hz was chosen to reduce the interference from neighboring notes. The passband was normalized by the Nyquist frequency (Fs / 2) and a 10th-order Butterworth bandpass filter was designed. For numerical stability, the filter was implemented in SOS

form using zp2sos() and applied with sosfilt(). The filtered output y(t) was then compared with the original signal x(t) in the time domain and using FFT magnitude spectra (Hann-windowed and normalized to the peak of the original spectrum).

Low-pass and high-pass filters were also created for the GUI applications. The filters were designed in a similar way to the band-pass filter, with a common cutoff frequency defined by using note pairs and taking the midpoint between their individual fundamental frequencies.

4) GUI Structure

A MATLAB App Designer GUI was implemented. When the “Load Audio File” button is pressed, the app reads the audio file, calculates and displays the sampling frequency, total duration, and note duration, and plots the original signal in both the time-domain and frequency-domain. The GUI also includes controls for filter type, filter order, and either a bandpass note or a cutoff note pair, depending on the selected filter type. The “Apply Filter” button designs the chosen filter, applies it to the full recording, and updates the filtered signal time-domain and spectrum plots. Finally, separate buttons were provided to play the original and filtered audio.

Results

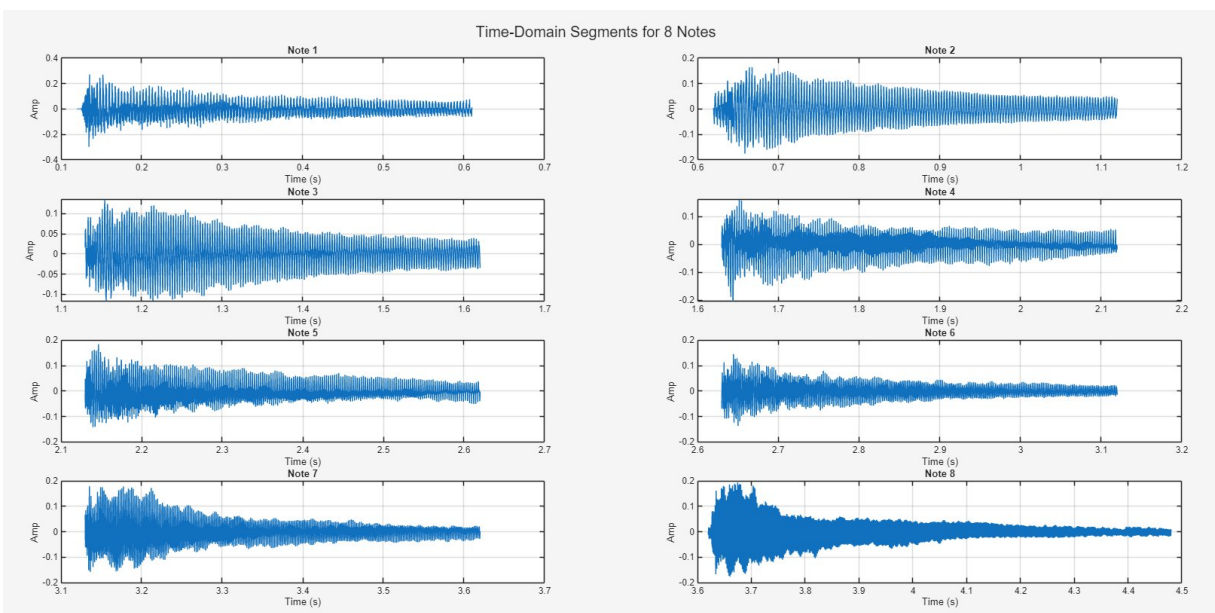

Figure 1: The 8 note segments in time domain

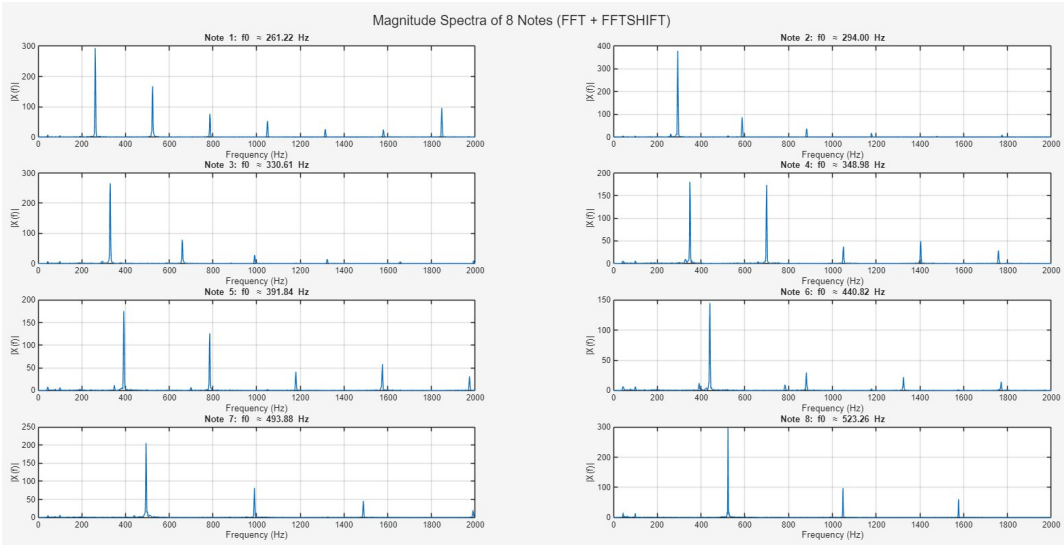

Figure 2: The 8 note segments in frequency domain

Note	Measured Frequency (Hz)	Theoretical Frequency (Hz)	% Error
C4	261.22	261.63	0.15
D4	294.00	293.66	0.12
E4	330.61	329.63	0.30
F4	348.98	349.23	0.07
G4	391.84	392.00	0.04
A4	440.82	440.00	0.19
B4	493.88	493.88	0.00
C5	523.26	523.25	0.002

Table 1: The measured vs theoretical fundamental frequency for each note

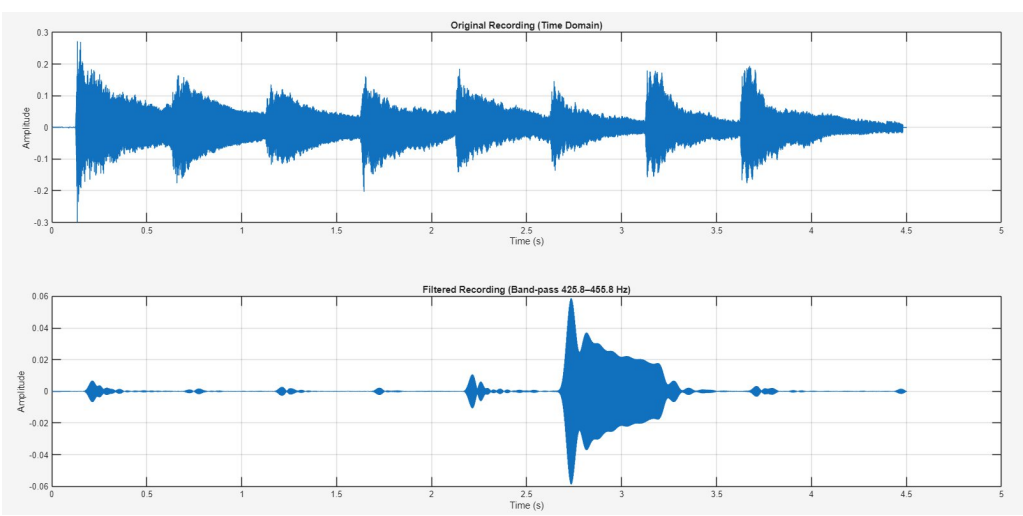

Figure 3: The original audio and the filtered audio in time domain

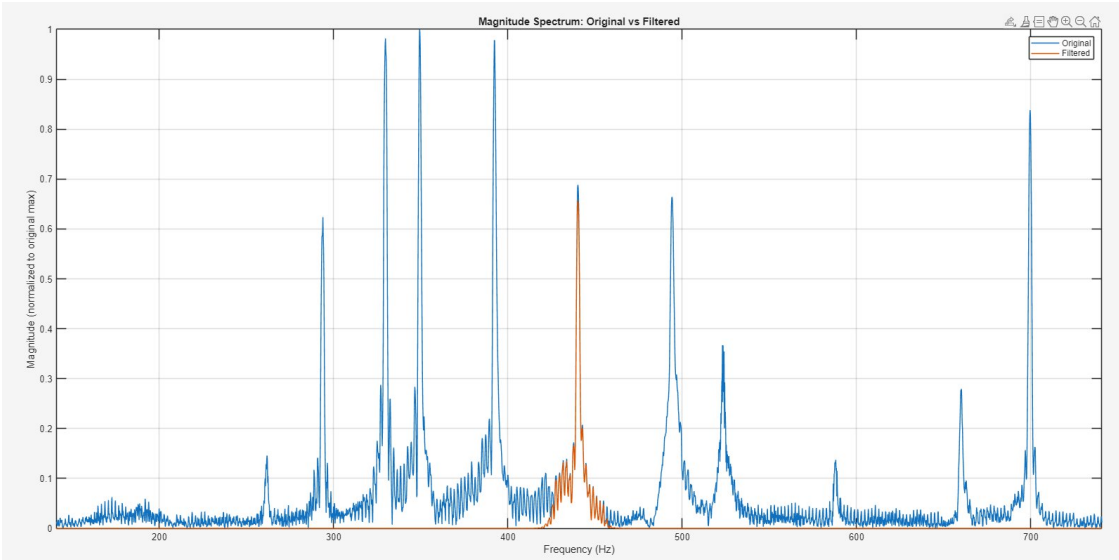

Figure 4: The original audio vs the filtered audio in frequency domain

Discussion & Conclusion

The estimated fundamental frequencies of the eight notes were close to the theoretical frequencies of the C major scale, and the percentage errors were small. The small differences are normal and can happen for a few reasons like segment length (when a note segment is short, the frequency resolution becomes worse), a real piano note might contain more than one strong peak, and the recording environment. A Hann window helps reduce spectral leakage, but it cannot remove it completely.

A Butterworth band-pass filter was chosen because it has a smooth passband and is easy to design. After filtering the full recording, the time-domain waveform showed that the signal became much weaker outside the target note region, since most other frequency components were removed. In the frequency domain, the filtered spectrum was mainly concentrated around the selected note's frequency band, and the other notes were largely reduced. Still, perfect isolation is not expected because nearby notes and harmonics can overlap in frequency.

In conclusion, this homework showed how to segment an audio signal, estimate note frequencies using the FFT, and design a digital filter to isolate one note. It also showed an important trade-off: using a narrower passband suppresses other notes better, but it can also reduce the target note if there are noise or small tuning differences.

Appendix

Moxey, J. (2025, December 23). *Music Note Fundamental frequencies - Songstuff*. Songstuff. https://www.songstuff.com/recording/article/music-fundamental-frequencies/#elementor-toc__heading-anchor-10

App Building OnRamp | Self-Paced Online Courses - MATLAB & Simulink. (n.d.). https://matlabacademy.mathworks.com/details/app-building-onramp/orab

Develop Apps Using App Designer. https://www.mathworks.com/help/matlab/app-designer.html